1. Pipeline（工作流程）
该研究的 pipeline 旨在将大语言模型（LLM）的生成能力转化为训练高效双编码器（Dual-Encoder）的监督信号
步骤 1：种子段落选择 (Seed Passage Selection) 从目标文档集中选择种子段落作为生成基础。研究发现随机采样 (Random Sampling) 在大多数情况下与复杂的局部主题选择效果相当，甚至能提供更好的多样性

步骤 2：多步查询生成 (Query Generation) 利用 LLM（如 Mixtral-8x7B-Instruct）通过 思维链（Chain-of-Thought, CoT）提示词生成查询
总结：产生 10 个词左右的主旨总结。
泛化：将总结改写为更通用的陈述，以确保查询不至于过窄。
提问：将陈述转化为问题格式。
精简：最后通过“搜索查询”提示词将上述结果精炼为简短、符合人类搜索习惯的查询

步骤 3：教师模型推理 (Teacher Inference Pipeline) 使用现有的 CLIR 系统（如 PLAID-X）检索出与生成查询最相关的 Top-50 文档，然后利用强大的 Cross-Encoder（如 Mono-mT5XXL） 对这些查询-文档对进行打分，产生“教师分数”

步骤 4：学生模型蒸馏 (Student Training) 以教师模型的分数分布为目标，利用 KL 散度损失函数 训练轻量级的学生双编码器（ColBERT-X），使其在保持检索效率的同时达到教师模型的精度

2. contribution
解决 CLIR 数据缺口：证明了合成查询在 CLIR 任务中可以达到甚至超过依赖机器翻译的 MS MARCO 数据集的训练效果

高效提示词设计：提出了针对检索任务优化的 CoT 提示词，通过“总结-泛化-提问”的路径生成高质量查询

降低对“相关性”的强假设：不同于以往方法（如 Promptagator）需要过滤掉无法检索到原文档的查询，G-D 仅假设集合中存在能回答该查询的文档即可，这降低了对生成模型微调的需求

3. 文章的思想
集合偏置 (Collection Bias) 优于通用性：该文章的核心思想是不追求产生一个通用的 CLIR 模型，而是通过利用目标集合本身的文档生成查询，使模型“偏向” (Bias)** 并适应特定的搜索集合和领域知识**

独立于黄金标准 (Independence from Gold Labels)：它假设 LLM 只要能产生一个“合理的搜索意图”，哪怕它与原始种子文档的联系不完美，只要教师模型能在庞大的语料库中找到与之匹配的答案，这个训练对就是有效的。这极大减少了对人工标注和高质量翻译的依赖

逻辑推理引导生成：通过思维链提示词模拟人类构建检索课题的过程（从理解主旨到抽象概念再到形成疑问），以此确保合成查询的语义深度和分布符合真实检索场景

简洁性至上：研究发现过于冗长或包含太多细节的查询（如 UMR 生成的）在训练中效果较差。因此，其思想核心之一是模拟真实用户的简短搜索行为，而非单纯追求生成文档的“逆向重写”

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SHORT-PAPER

Generate-Distill: Training Cross-Language IR Models with Synthetically-Generated Data

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Published: 13 July 2025

Citation in BibTeX format

SIGIR '25: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval
July 13 - 18, 2025
Padua, Italy

Conference Sponsors:
SIGIR

SIGIR '25: Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (July 2025)
https://doi.org/10.1145/3726302.3730201
ISBN: 9798400715921

Generate-Distill: Training Cross-Language IR Models with Synthetically-Generated Data

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Abstract

Most pretrained language models that support neural information retrieval are fine-tuned on the MS MARCO dataset. MS MARCO is expressed in English, so it naturally supports monolingual English retrieval. However, for Cross-Language Information Retrieval (CLIR), no similar training data naturally exists that matches the languages of the query and the documents. The main ways to address this problem have been to continue to fine-tune with English data, or to translate MS MARCO queries and/or documents to match the CLIR setting. Machine translation often introduces errors that reduce retrieval effectiveness. It is usually easy to find target language documents suitable for training, but difficult to find naturally-occurring queries in the query language. An alternative is to train on naturally-occurring documents and synthetically-generated queries. Generate-Distill uses this approach with state-of-the-art distillation methods to match the effectiveness of training with translated MS MARCO across different domains.

CCS Concepts

• **Information systems** → **Search engine indexing**; **Language models**; *Multilingual and cross-lingual retrieval.*

Keywords

CLIR, Dense retrieval, distillation, Translate-Train, ColBERT

ACM Reference Format:
Dawn Lawrie, Efsun Kayi, Eugene Yang, James Mayfield, Douglas W. Oard, and Scott Miller. 2025. Generate-Distill: Training Cross-Language IR Models with Synthetically-Generated Data. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25), July 13–18, 2025, Padua, Italy.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3726302.3730201

1 Introduction

Recent advances in training state-of-the-art dual encoders have found improvements by using distillation from score distributions rather than using a contrastive loss between a positive and a negative example. Training a dense retrieval model with distillation requires queries and scores for a set of passages which, when used for Cross-Language Information Retrieval (CLIR), should match the query and document languages. For instance, to train a model that can rank Chinese documents with English queries, one would train the dual encoder with English queries and Chinese passages.

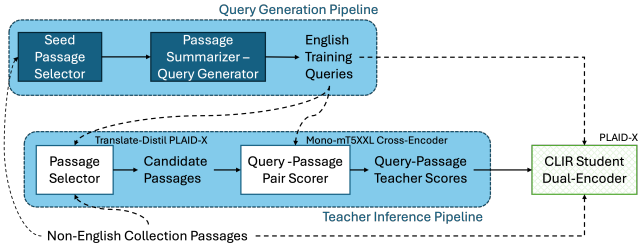

Figure 1: Generate-Distill training pipeline. Queries are synthesized based on a selected passage. The queries are used for distillation, where the white boxes are the fixed teacher models. The passage selector is a current CLIR system.

Resources with large sets of realistic queries are rare. For English, MS MARCO [27] is a popular and arguably robust choice. It is possible to use MS MARCO for CLIR by translating the queries, passages, or both to match the CLIR setting. Translation introduces errors, however, which Translate-Distill [37] partially overcomes by selecting and scoring passages using monolingual English retrieval. However, this approach is **limited by its reliance on effective machine translation, and also in some applications by limitations of the MS MARCO license (which only allows research but not commercial use).** Those limitations motivate our interest in whether machine-generated queries can be used to effectively train dense CLIR retrieval models. We assume that a document collection exists as a source of passages. This is generally true, since otherwise there would be nothing for a CLIR system to search. If English were the query language and given some collection of documents, MS MARCO queries might be used. However, the collection would then need content matching those queries. The alternative we explore in this paper is to use a generative Large Language Model (LLM) to generate suitable queries.

Specifically, we explore whether LLMs can create synthetic queries that are sufficiently similar to actual future queries to train an effective dense CLIR retrieval model. Figure 1 outlines our proposed training process, Generate-Distill (G-D). G-D shares the same Teacher Inference Pipeline in Figure 1 with Translate-Distill; however, input into that pipeline is completely different. Instead of MS MARCO queries and translated MS MARCO passages, G-D uses passages in the document collection and synthetically generated

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SIGIR '25, Padua, Italy
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ACM ISBN 979-8-4007-1592-1/2025/07
https://doi.org/10.1145/3726302.3730201

queries. The goal of this work is not to produce generic CLIR retrieval models but instead to bias the retrieval model to the search collection. While synthetic query generation is not new [11, 24], we focus on how passages from which queries are generated are selected, and on how prompts are constructed. We find that both factors influence the effectiveness of the trained model. Experiments on the TREC NeuCLIR 2022 [18], 2023 [19] and 2024 [20] news test collections show that this strategy can be as effective as training on translated MS MARCO. We observe similar effects on domain-specific CLIR tasks using NeuCLIR 2023 and 2024 Technical Document collection and HC3 Chinese Tweets collection [21].

2 Background and Related Work

Using generation in a retrieval pipeline has become increasingly popular as LLMs become more capable. Early approaches explored query generation for document expansion [4, 28]. This was followed by neural variants, such as those of Dong et al. [9] that learn an interaction between a pseudo-query vector and a passage; and of Zeng et al. [38], which creates an augmented corpus representation based on LLM-generated synthetic questions and keywords.

Generating queries that retrieve specific documents is one way to create positive examples with which to train a ranker. For example, AugTriever [25] uses title and anchor text extraction, salient span selection, summarization, and keyphrase generation. An advantage of the approach is its ability to produce training examples in new domains [17, 23, 29] and languages [30, 32, 39]. There is also evidence that generated queries have lower retrievability bias (preference for some document classes) than human-generated queries [1]. Another use of generated queries, exemplified by Contriever [12] and Promptagator [8], is to contrastively train language models. Recently more nuanced approaches to passage selection were introduced monolingually by DUQGEN [2] and translingually by JH POLO [24]. These approaches assume that a query is related to its source passage, so they remove queries for which a search did not return the source document at a high rank. For instance, JH POLO preselects positive and negative passages, then has an LLM generate a query to differentiate them. Our Generate-Distill approach replaces this with a less restrictive assumption that the collection contains documents relevant to the query.

The work most closely related to ours is Unsupervised Multilingual dense Retriever (UMR) [11]. UMR trains a multilingual Dense Passage Retriever (mDPR) [14] using English documents, generated non-English queries, the language modeling-adapted variant of the 3 billion parameter mT5-XL [36], and distillation of query-passage scores. Our work differs in prompt design and model architecture. For example, Huang et al. [11] use the simple prompt "Based on the passage, please write a question in L," where L is a language. We found, however, that a Chain-of-Thought prompt [33] was required to generate queries that were useful for training a state-of-the-art dual-encoder. UMR generally underperforms DPR-MT, which prior work [15, 19] has shown to be, in turn, significantly worse than Translate-Distill [37]. We therefore principally compare to a state-of-the-art Translate-Distill baseline in our experiments.

Given the document, <doc>
Step 1: produce a summary of the main concept in ten English words.
Step 2: rewrite the summary as a more general statement.
Step 3: rephrase the general statement as a question.
Output the response as a json object using the general statement format
`{"summary": <summary>, "rewrite": <rewrite>, "question": <question>}`

Figure 2: Chain-of-Thought prompt for query generation.

Given the following summaries of a document <summary> and <rewrite>, write a search query that begins "Find information about" such that the summarized document is relevant to me in 20 or fewer words. Be sure that the search query mentions any named entities in the summary.

Figure 3: Final "Search Query" prompt.

3 Query Generation

This work uses LLMs to create training queries for neural CLIR retrieval models. When training such a model, training queries and passages should ideally approximate distributions that the model will see at inference time. The number of queries required can vary with the domain of the collection, the language pair, the pretrained language model (which may model some languages better than others), and the architecture of the retrieval model. In this work, we target the creation of about half a million queries. If the collection is larger than the desired number of training examples, a selection process is needed. Once the passages are selected, we use a generative model to create training queries. For reproducibility, we chose the open-source `Mixtral-8x7B-Instruct`.[1]

Figure 2 shows the first three steps in our query generation process, implemented using a single Chain-of-Thought prompt [16] that mimics roughly how a person might develop a topic for an information retrieval test collection. First, to focus on the central concepts in the passage, the model inspects (and implicitly translates) the passage, summarizing it using several English words. Second, to encourage the generation of queries that are not overly narrow, the model generalizes from that summary, creating a statement that other passages may also address. Finally, the model reformulates the statement as a question. Many of the original summaries are rather long, so the rewrite frequently adds additional details from the original passage, contrary to our intent. Therefore, we also ask the same LLM to further refine the output of its Chain-of-Thought prompt, using the prompt shown in Figure 3. After light processing to remove any unrequested explanations added by the LLM, this yielded more concise queries.[2] For the experiments in this paper, we used as training queries the output of the first (Summary) step from the Chain of Thought prompt and from our last ("Search Query") prompt.

Generate-Distill does not have the requirement of most prior work to remove generated queries for which the seed passage is

[1] https://huggingface.co/mistralai/Mixtral-8x7B-Instruct-v0.1
[2] Our processing scripts and generated data are available at https://huggingface.co/datasets/hltcoe/Generate-Distill-Data.

Table 1: Collection statistics.

	NeuCLIR News			TechDocs	HC3
	Chinese	Persian	Russian	Chinese	Chinese
# Docs	3.2M	2.2M	4.6M	396k	5.6M
# Passages	5.2M	3.7M	6.8M	399k	5.6M
# Topics	167	174	171	111	50
# Seed passage:					
Full collection	–	–	–	396k	–
Random sample	556k	556k	554k	–	400k
Passage quality	500k	500k	500k	–	–
Topical selection	–	–	–	–	253k

not ranked highly [8]. Such filtering requires the generative model to capture not just the query language model, but also the concept of relevance; the latter requires more tuning [34]. Here, we only need generative models to provide a reasonable query, regardless of its relationship to the seed passage from which it was constructed. This choice makes us less concerned with hallucinations from the LLM since we merely assume the existence of at least one passage that satisfies the query. Seed passages serve only to ground the generation in a plausible topic for the collection.

4 Experiment Design

Test Collections. We use three CLIR evaluation suites: NeuCLIR News [18–20], NeuCLIR Technical Documents [19, 20], and HC3 Chinese Twitter [21]. NeuCLIR News contains Chinese, Persian and Russian articles from CommonCrawl News and topics from TREC 2022, 2023, and 2024. NeuCLIR Technical Documents (TechDocs) contains Chinese academic paper abstracts from the CSL collection [22] and topics from TREC 2023 and 2024. HC3 Chinese contains Tweet threads, the vast majority of which are either quite short or not topically focused, and a set of topics. In each case we use English Title+Description as evaluation queries. Table 1 summarizes the collection statistics. To compare the effects of *different languages, while controlling genre* we use the NeuCLIR News collections in Chinese, Persian, and Russian. To study effects of *fixing the document language and varying the genre*, we compare results on NeuCLIR Chinese News to results on NeuCLIR (Chinese) Technical Documents and HC3 Chinese.

Query Generation. Our main experiments are conducted using the results of our final "Search Query" prompt from Figure 3. Since the collection sizes vary, we use collection-specific approaches to select the seed passages on which to base query generation. The NeuCLIR News collections are far larger than the half million or so seed passages that we need for our experiments. We tried two selection techniques, one by random sampling and the other by selecting the half million passages that achieve the highest passage quality estimation score [3].[3] The NeuCLIR Technical Documents test collection is smaller than our target size for the number of seed passages; for it we use every passage in the collection as a seed passages. Prior work has shown that most of the Tweet threads in the HC3 collection lack clear topical focus [24], so for the HC3 Chinese collection we tried a topic-guided selection approach. Following Mayfield et al. [24], we found the 100 top-ranked tweet threads in HC3 Chinese by using a Chinese machine translation of a summary from the WCEP multi-document summarization dataset [10] as a BM25 query. We repeated this for 10,200 WCEP summaries and then removed duplicates from the resulting set. This resulted in 252,730 (mostly topical) Chinese Tweet threads that we use as seed passages from which to generate training queries for that collection. We compare this to random selection. The bottom of Table 1 shows the number of seed passages that these approaches produce.[4]

Retrieval Model. Following Nair et al. [26], we create passages from the documents using a sliding window of 180 tokens and a stride of 90. For each generated query, we first identify the top fifty passages using the PLAID-X engine [37] with a model trained using the Translate-Distill pipeline[5], and then use an existing Mono-mT5XXL [13] cross-encoder to obtain teacher scores. Our ColBERT-X dual encoder student model is initialized with XLM-RoBERTa-Large [5], then fine-tuned for 200,000 gradient steps with a per-GPU batch size of eight queries, with six randomly selected associated passages for each. We train with 8 V100s using an Adam optimizer with a learning rate 5×10^{-6} on KL-divergence loss, which is similar to the training process introduced in Translate-Distill [37].

After training, we encode and index the documents with the trained ColBERT-X model using the PLAID-X implementation [37], which is a public CLIR fork of PLAID [31]. During PLAID-X encoding, we partition each document into 450-token passages without overlap. While this deviates from the common 180-token stride-90 segmentation, increasing the passage size generally improves retrieval effectiveness and reduces the index size (because with no overlap it never indexes any token twice). All models trained in this work and our generated queries are available on Huggingface.[6]

Evaluation. For evaluation, document scores are aggregated by taking the maximum score among its passages [7]. We report nDCG@20, which was the primary metric for each collection when it was introduced. We test for significant *differences* using a paired t-test with multiple testing correction, and for *equivalence* within a predefined tolerance of ±0.05 using the Two One-Sided t-Tests (TOST) equivalence test. For TOST, the *null hypothesis* is that the difference between the means of two samples of values are *larger* than a specified tolerance (±0.05 herein), which tests for the opposite of paired t-tests. A low p-value in TOST means that the probability of the difference spanning beyond the tolerance is low, which rejects the null hypothesis and indicates that the true difference is likely to be smaller than the tolerance. In the next section, we provide the p-values of t-tests when claiming differences and p-values of TOST when claiming *no meaningful differences* (i.e., differences within ±0.05).

5 Results

[3]https://huggingface.co/pyterrier-quality/mqt5-small

[4]We generally targeted about 500k training queries, but the specific numbers vary a bit because we ran these experiments at different times.

[5]We used models from https://huggingface.co/hltcoe/plaidx-large-LANG-tdist-mt5xxl-engeng where LANG is zho, fas, or rus for Chinese, Persian, or Russian model, respectively.

[6]https://huggingface.co/collections/hltcoe/generate-distill-680fe91eccaf0e31564fe92a

Table 2: nDGC@20, "Search Query" prompt. Number of Generate-Distill training queries in parentheses. MS MARCO Sample approximates that number (NeuCLIR News: 500k, NeuCLIR Tech: 300k, HC3: 400k). Generate-Distill doc selection: full-collection (TechDocs), random sample (otherwise).

	NeuCLIR News			NeuCLIR	HC3
	Persian (473K)	Russian (498K)	Chinese (528K)	TechDocs (296K)	Twitter (385K)
Baselines					
DT » BM25+RM3	0.404	0.355	0.396	0.235	0.262
ColBERT-X (TT)	0.502	0.465	0.458	0.286	0.397
ColBERT-X with Translate-Distill (using the specified training data)					
Full MS MARCO	**0.572**	0.534	0.520	**0.365**	0.472
MS MARCO Sample	0.560	**0.537**	0.527	0.343	0.434
Generate-Distill	0.536	0.508	**0.532**	0.346	**0.478**

Generate-Distill works as well as Translate-Distill for Chinese. Table 2 shows results for Generate-Distill using the queries from the "Search Query" prompt in Figure 3. With this prompt, Generate-Distill results are not meaningfully different on Chinese NeuCLIR News from results of Translate-Distill trained using all of MS MARCO (TOST $p = 0.003$). That is not true for Persian or Russian, where Translate-Distill is significantly better than Generate-Distill (t-test $p = 0.008$ and $p = 0.026$ respectively). Inspection of generated queries reveals no obvious systematic differences for Persian or Russian. We note, however, that Yang et al. [37] did find that when using a CLIR teacher model to score passages (rather than using English scores, as Translate-Distill does), results were clearly language-dependent, which may be overcome by using newer rerankers, such as Rank-1 [35], as the teacher.

"MS MARCO Sample" in Table 2 denotes a stratified sample that reflects the observed variation in document scores in the sample.[7] Using a sample size that roughly matches the number of generated queries yields about the same nDCG@20 as using all of MS MARCO. These diminishing returns from additional training suggest that 500k training queries are more than enough, at least if the training set is of sufficiently high quality. As for other CLIR baselines, ColBERT-X Translate-Train (TT)[26] and BM25+RM3 (over machine-translated documents)[6] are both significantly worse than Generate-Distill in all three languages (t-test after multi-testing correction over five collections, $p = 0.033$ and $p < 0.0001$)

Considering other genres, Generate-Distill is not meaningfully different from Translate-Distill for Technical Documents when Translate-Distill is trained on Full MS MARCO, or on MS MARCO Sample (TOST $p = 0.029$ and $p = 0.004$, respectively). Likewise, evaluating on HC3 demonstrates Generate-Distill has similar effectiveness with Translate-Distill on Full MS MARCO (TOST $p = 0.035$), while comparing Generate-Distill with Translate-Distill on MS MARCO Sample is inconclusive.

Choice of Query. Table 3 shows the effect of using the Summary from the first part of the Chain-of-Thought prompt in Figure 2 rather than the result of the "Search Query" prompt. To avoid language effect confounds, we focus here on Chinese collections.

[7]Stratification is on the distribution of document scores per query at percentiles 1/5/10/25/50/75/90/95/99. Queries binned based on their document score distribution; sample drawn from each bin, proportional to number of queries in that bin.

Table 3: Query and selection effects, Chinese, nDCG@20.

Query Prompt	Seed Selection	News	TechDocs	Twitter
Search Query	Full collection	–	**0.346**	–
Search Query	Random sample	**0.532**	–	**0.478**
Search Query	Passage quality	0.500	–	–
Search Query	Topical selection	–	–	0.462
Summary	Full collection	–	0.339	–
Summary	Random sample	0.517	—	0.412
UMR	Random sample	0.456	–	–

For Chinese News and Technical Documents, those two ways of selecting queries yielded no meaningful difference (TOST $p = 0.001$, $p < 0.0001$, respectively), but for HC3, the difference was weakly significant (t-test $p = 0.06$). We also tried the prompt used by Huang et al. [11] for their UMR method on Chinese NeuCLIR News, finding that training on the resulting queries yielded results that were statistically significantly worse than those achieved when training on queries generated by either our "Search Query" prompt or the Summary from the Chain-of-Thought prompt (t-test $p < 0.0001$ and $p = 0.0005$, respectively). An inspection of the queries generated by the UMR prompt found them to be much longer than those generated by "Search Query" prompt and to include many details, thus appearing to us less similar to the way we would expect people to actually craft queries. We note, however, that our LLM differs from theirs, and that we trained ColBERT-X rather than mDPR. So what we can see here is just the effect of the difference in prompts.

Seed Passage Selection. Table 3 also shows results for different ways of selecting seed passages. The random sampling (or, for Technical Documents, full collection) approach used in Table 2 leads to statistically significantly better effectiveness than using passage quality for Chinese News (t-test $p = 0.006$). Inspection of passages with high passage quality indicates that pop culture and weather passages scored highly by that measure, but few topics in the NeuCLIR Chinese News test collection focus on those issues. Interestingly, there is weak evidence that there may be no meaningful difference between the use of random sampling or topical selection for Chinese Twitter (TOST $p = 0.063$, slightly above our 0.05 significance level). This suggests that our rather complex topical selection approach, using documents from a side collection as queries, may not be worth the effort. We suspect that topical selection limits the diversity of the training set.

6 Conclusion and Future Work

We have explored the ability of LLMs to generate query sets that can be used to train efficient dual-encoder models for effective CLIR. We find that random sampling works at least as well as more complex techniques for selecting seed passages from which to generate queries, and that the design of the LLM prompts used to generate the training queries can affect the quality of the resulting CLIR model. Future work will explore whether alternative teacher passage selection and scorer approaches can improve effectiveness.

References

[1] Amin Abolghasemi, Suzan Verberne, Arian Askari, and Leif Azzopardi. 2023. Retrievability Bias Estimation using Synthetically Generated Queries. In *Proceedings of the 32nd ACM Int'l Conference on Information and Knowledge Management*. 3712–3716.

[2] Ramraj Chandradevan, Kaustubh D Dhole, and Eugene Agichtein. 2024. DUQGen: Effective Unsupervised Domain Adaptation of Neural Rankers by Diversifying Synthetic Query Generation. *arXiv preprint arXiv:2404.02489* (2024).

[3] Xuejun Chang, Debabrata Mishra, Craig Macdonald, and Sean MacAvaney. 2024. Neural Passage Quality Estimation for Static Pruning. In *Proceedings of the 47th Int'l ACM SIGIR Conference on Research and Development in Information Retrieval*. 174–185.

[4] David R. Cheriton. 2019. From doc2query to docTTTTTquery. https://api.semanticscholar.org/CorpusID:208612557

[5] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised Cross-lingual Representation Learning at Scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Online, 8440–8451.

[6] Cash Costello, Eugene Yang, Dawn Lawrie, and James Mayfield. 2022. Patapsco: A Python Framework for Cross-Language Information Retrieval Experiments. In *Proceedings of the 44th European Conference on Information Retrieval (ECIR)*.

[7] Zhuyun Dai and Jamie Callan. 2019. Deeper Text Understanding for IR with Contextual Neural Language Modeling. In *Proceedings of the 42nd Int'l ACM SIGIR Conference on Research and Development in Information Retrieval*. 985–988.

[8] Zhuyun Dai, Vincent Y. Zhao, Ji Ma, Yi Luan, Jianmo Ni, Jing Lu, Anton Bakalov, Kelvin Guu, Keith B. Hall, and Ming-Wei Chang. 2022. Promptagator: Few-shot Dense Retrieval From 8 Examples. arXiv:2209.11755 [cs.CL] https://arxiv.org/abs/2209.11755

[9] Qian Dong, Yiding Liu, Qingyao Ai, Haitao Li, Shuaiqiang Wang, Yiqun Liu, Dawei Yin, and Shaoping Ma. 2023. I3 Retriever: Incorporating Implicit Interaction in Pre-trained Language Models for Passage Retrieval. In *Proceedings of the 32nd ACM Int'l Conference on Information and Knowledge Management* (Birmingham, United Kingdom) *(CIKM '23)*. Association for Computing Machinery, New York, NY, USA, 441–451. https://doi.org/10.1145/3583780.3614923

[10] Demian Gholipour Ghalandari, Chris Hokamp, Nghia The Pham, John Glover, and Georgiana Ifrim. 2020. A Large-Scale Multi-Document Summarization Dataset from the Wikipedia Current Events Portal. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Online, 1302–1308.

[11] Chao-Wei Huang, Chen-An Li, Tsu-Yuan Hsu, Chen-Yu Hsu, and Yun-Nung Chen. 2024. Unsupervised Multilingual Dense Retrieval via Generative Pseudo Labeling. In *Findings of the Association for Computational Linguistics: EACL 2024*. 736–746.

[12] Gautier Izacard, Mathilde Caron, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. 2022. Unsupervised Dense Information Retrieval with Contrastive Learning. *Transactions on Machine Learning Research* (2022). https://openreview.net/forum?id=jKN1pXi7b0

[13] Vitor Jeronymo, Roberto Lotufo, and Rodrigo Nogueira. 2023. NeuralMind-UNICAMP at 2022 TREC NeuCLIR: Large Boring Rerankers for Cross-lingual Retrieval. *arXiv preprint arXiv:2303.16145* (2023).

[14] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 6769–6781.

[15] Omar Khattab and Matei Zaharia. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. In *Proceedings of the 43rd Int'l ACM SIGIR conference on research and development in Information Retrieval*. 39–48.

[16] Takeshi Kojima, Shixiang (Shane) Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large Language Models are Zero-Shot Reasoners. In *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35. Curran Associates, Inc., 22199–22213. https://proceedings.neurips.cc/paper_files/paper/2022/file/8bb0d291acd4acf06ef112099c16f326-Paper-Conference.pdf

[17] Thiago Laitz, Konstantinos Papakostas, Roberto Lotufo, and Rodrigo Nogueira. 2024. InRanker: Distilled Rankers for Zero-shot Information Retrieval. arXiv:2401.06910 [cs.IR]

[18] Dawn Lawrie, Sean MacAvaney, James Mayfield, Paul McNamee, Douglas W. Oard, Luca Soldanini, and Eugene Yang. 2023. Overview of the TREC 2022 NeuCLIR Track. In *The Thirty-First Text REtrieval Conference (TREC 2022) Proceedings*.

[19] Dawn Lawrie, Sean MacAvaney, James Mayfield, Paul McNamee, Douglas W. Oard, Luca Soldanini, and Eugene Yang. 2024. Overview of the TREC 2023 NeuCLIR Track. In *The Thirty-Second Text REtrieval Conference (TREC 2023) Proceedings*.

[20] Dawn Lawrie, Sean MacAvaney, James Mayfield, Paul McNamee, Douglas W. Oard, Luca Soldanini, and Eugene Yang. 2025. Overview of the TREC 2024 NeuCLIR Track. In *The Thirty-Third Text REtrieval Conference (TREC 2024) Proceedings*.

[21] Dawn Lawrie, James Mayfield, Douglas W Oard, Eugene Yang, Suraj Nair, and Petra Galuščáková. 2023. HC3: A Suite of Test Collections for CLIR Evaluation over Informal Text. In *Proceedings of the 46th Int'l ACM SIGIR Conference on Research and Development in Information Retrieval (Taipei, Taiwan) (SIGIR'23)*.

[22] Yudong Li, Yuqing Zhang, Zhe Zhao, Linlin Shen, Weijie Liu, Weiquan Mao, and Hui Zhang. 2022. CSL: A Large-scale Chinese Scientific Literature Dataset. In *Proceedings of the 29th Int'l Conference on Computational Linguistics*. Int'l Committee on Computational Linguistics, Gyeongju, Republic of Korea, 3917–3923. https://aclanthology.org/2022.coling-1.344

[23] Ji Ma, Ivan Korotkov, Yinfei Yang, Keith Hall, and Ryan McDonald. 2020. Zero-Shot Neural Passage Retrieval via Domain-Targeted Synthetic Question Generation. *arXiv preprint arXiv:2004.14503* (2020).

[24] James Mayfield, Eugene Yang, Dawn Lawrie, Samuel Barham, Orion Weller, Marc Mason, Suraj Nair, and Scott Miller. 2023. Synthetic Cross-Language Information Retrieval Training Data. *arXiv preprint arXiv:2305.00331* (2023).

[25] Rui Meng, Ye Liu, Semih Yavuz, Divyansh Agarwal, Lifu Tu, Ning Yu, Jianguo Zhang, Meghana Bhat, and Yingbo Zhou. 2024. AugTriever: Unsupervised Dense Retrieval by Scalable Data Augmentation. In *DCAI24 workshop at CIKM2024*.

[26] Suraj Nair, Eugene Yang, Dawn Lawrie, Kevin Duh, Paul McNamee, Kenton Murray, James Mayfield, and Douglas W. Oard. 2022. Transfer Learning Approaches for Building Cross-Language Dense Retrieval Models. In *Advances in Information Retrieval: 44th European Conference on IR Research, ECIR 2022, Stavanger, Norway, April 10–14, 2022, Proceedings, Part I* (Stavanger, Norway). Springer-Verlag, Berlin, Heidelberg, 382–396.

[27] Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. 2016. MS MARCO: A Human Generated MAchine Reading COmprehension Dataset. *arXiv preprint arXiv:1611.09268* (2016). arXiv:1611.09268 http://arxiv.org/abs/1611.09268

[28] Rodrigo Nogueira, Wei Yang, Jimmy Lin, and Kyunghyun Cho. 2019. Document Expansion by Query Prediction. arXiv:1904.08375 [cs.IR] https://arxiv.org/abs/1904.08375

[29] Revanth Gangi Reddy, Vikas Yadav, Md Arafat Sultan, Martin Franz, Vittorio Castelli, Heng Ji, and Avirup Sil. 2022. Towards Robust Neural Retrieval with Source Domain Synthetic Pre-Finetuning. In *Proceedings of the 29th Int'l Conference on Computational Linguistics*. 1065–1070.

[30] Houxing Ren, Linjun Shou, Ning Wu, Ming Gong, and Daxin Jiang. 2022. Empowering Dual-Encoder with Query Generator for Cross-Lingual Dense Retrieval. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 3107–3121.

[31] Keshav Santhanam, Omar Khattab, Christopher Potts, and Matei Zaharia. 2022. PLAID: An Efficient Engine for Late Interaction Retrieval. In *Proceedings of the 31st ACM Int'l Conference on Information & Knowledge Management*. 1747–1756.

[32] Nandan Thakur, Jianmo Ni, Gustavo Hernandez Abrego, John Wieting, Jimmy Lin, and Daniel Cer. 2024. Leveraging LLMs for Synthesizing Training Data Across Many Languages in Multilingual Dense Retrieval. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Kevin Duh, Helena Gomez, and Steven Bethard (Eds.). Association for Computational Linguistics, Mexico City, Mexico, 7699–7724. https://doi.org/10.18653/v1/2024.naacl-long.426

[33] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed H. Chi, Quoc Le, and Denny Zhou. 2022. Chain of Thought Prompting Elicits Reasoning in Large Language Models. *CoRR* abs/2201.11903 (2022). arXiv:2201.11903 https://arxiv.org/abs/2201.11903

[34] Orion Weller, Benjamin Van Durme, Dawn Lawrie, Ashwin Paranjape, Yuhao Zhang, and Jack Hessel. 2025. Promptriever: Instruction-Trained Retrievers Can Be Prompted Like Language Models. In *Proceedings of the Thirteenth Int'l Conference on Learning Representations*. Singapore. https://arxiv.org/abs/2409.11136

[35] Orion Weller, Kathryn Ricci, Eugene Yang, Andrew Yates, Dawn Lawrie, and Benjamin Van Durme. 2025. Rank1: Test-Time Compute for Reranking in Information Retrieval. arXiv:2502.18418 [cs.IR] https://arxiv.org/abs/2502.18418

[36] Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. mT5: A Massively Multilingual Pre-trained Text-to-Text Transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Online, 483–498. https://doi.org/10.18653/v1/2021.naacl-main.41

[37] Eugene Yang, Dawn Lawrie, James Mayfield, Douglas W Oard, and Scott Miller. 2024. Translate-Distill: Learning Cross-Language Dense Retrieval by Translation and Distillation. In *European Conference on Information Retrieval*. Springer, 50–65.

[38] Qiuhai Zeng, Zimeng Qiu, Dae Yon Hwang, Xin He, and William M. Campbell. 2024. Unsupervised Text Representation Learning via Instruction-Tuning for Zero-Shot Dense Retrieval. In *DCAI24 workshop at CIKM2024*.

[39] Shengyao Zhuang, Linjun Shou, and Guido Zuccon. 2023. Augmenting Passage Representations with Guery Generation for Enhanced Cross-Lingual Dense Retrieval. In *Proceedings of the 46th Int'l ACM SIGIR Conference on Research and Development in Information Retrieval*. 1827–1832.